

EES301: Global biogeochemical cycles and Climate change

The Sulfur Cycle

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EES301

Introduction

- The live biosphere contains relatively little sulfur. Major global pools of “S” are found in **biogenic pyrite, seawater, and evaporites** derived from ocean water.
- “S” cycling involves both **abiotic and biotic transformations**
- “S” is **highly redox active**, used in microbial respiratory processes as both an electron donor and an acceptor in different environments (like N compounds).
- No sulfur gas is a **long-lived or major constituent in the atmosphere**.
- **Sulfate is the second most abundant anion** in rivers (HCO_3^-) and seawater (Cl^-).
- Understanding the biogeochemistry of S is of economic significance – **Many metals are mined from sulfide minerals**.

Major S reservoirs

Reservoir	Major forms of sulfur	Mass (Tg S) ^b
Atmosphere	Gas: H ₂ S, SO ₂ , DMS, OCS Aerosols: H ₂ SO ₄ , (NH ₄) ₂ SO ₄	4.8
Pedosphere and biosphere	Sulfates, organic S, pyrite	3×10^5
Rivers and lakes	SO ₄ ²⁻	300
Oceans	SO ₄ ²⁻	1.3×10^9
Marine sediments	Pyrite, gypsum, anhydrite	3×10^8
Lithosphere	Gypsum, anhydrite, metal sulfides, sulfur	2×10^{10}

^bTg is 10¹² g

DMS: Dimethyl sulfide, OCS: Carbonyl sulfide

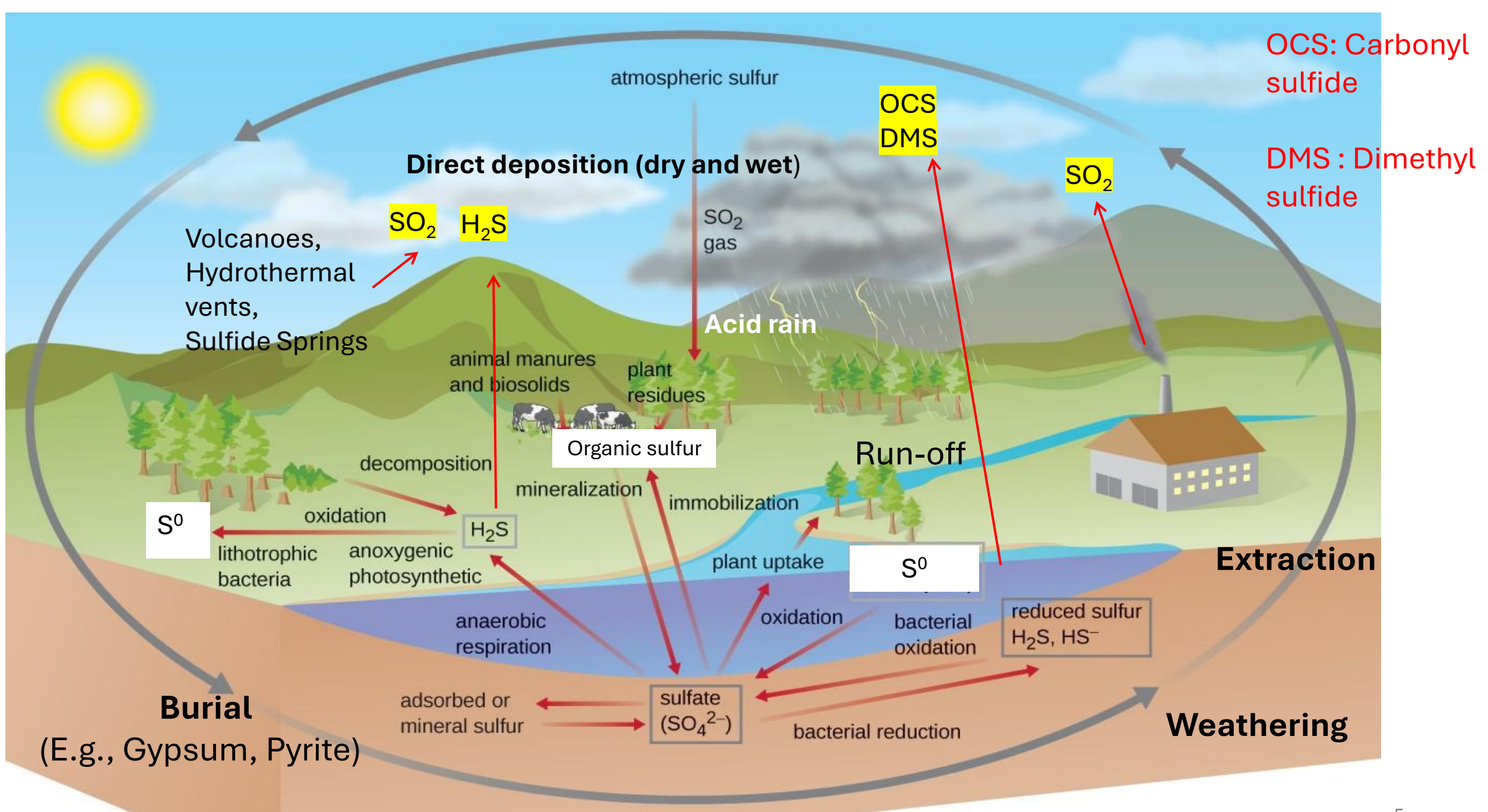
Compound	Formula	Oxidation state(s) of sulfur
Sulfide	S^{2-}	-2
Sulfur ^a	S_8^0	0
Hyposulfite (dithionite)	$S_2O_4^{2-}$	+2
Sulfite	SO_3^{2-}	+4
Thiosulfate ^b	$S_2O_3^{2-}$	0, +4
Dithionate	$S_2O_6^{2-}$	+4
Trithionate	$S_3O_6^{2-}$	0, +4
Tetrathionate	$S_4O_6^{2-}$	0, +4
Pentathionate	S_5O_6	0, +4
Sulfate	SO_4^{2-}	+6

^a Occurs in an octagonal ring in crystalline form.

^b Outer S has a valence of 0; inner S has a valence of +4. *Source: Ehrlich, 1981*

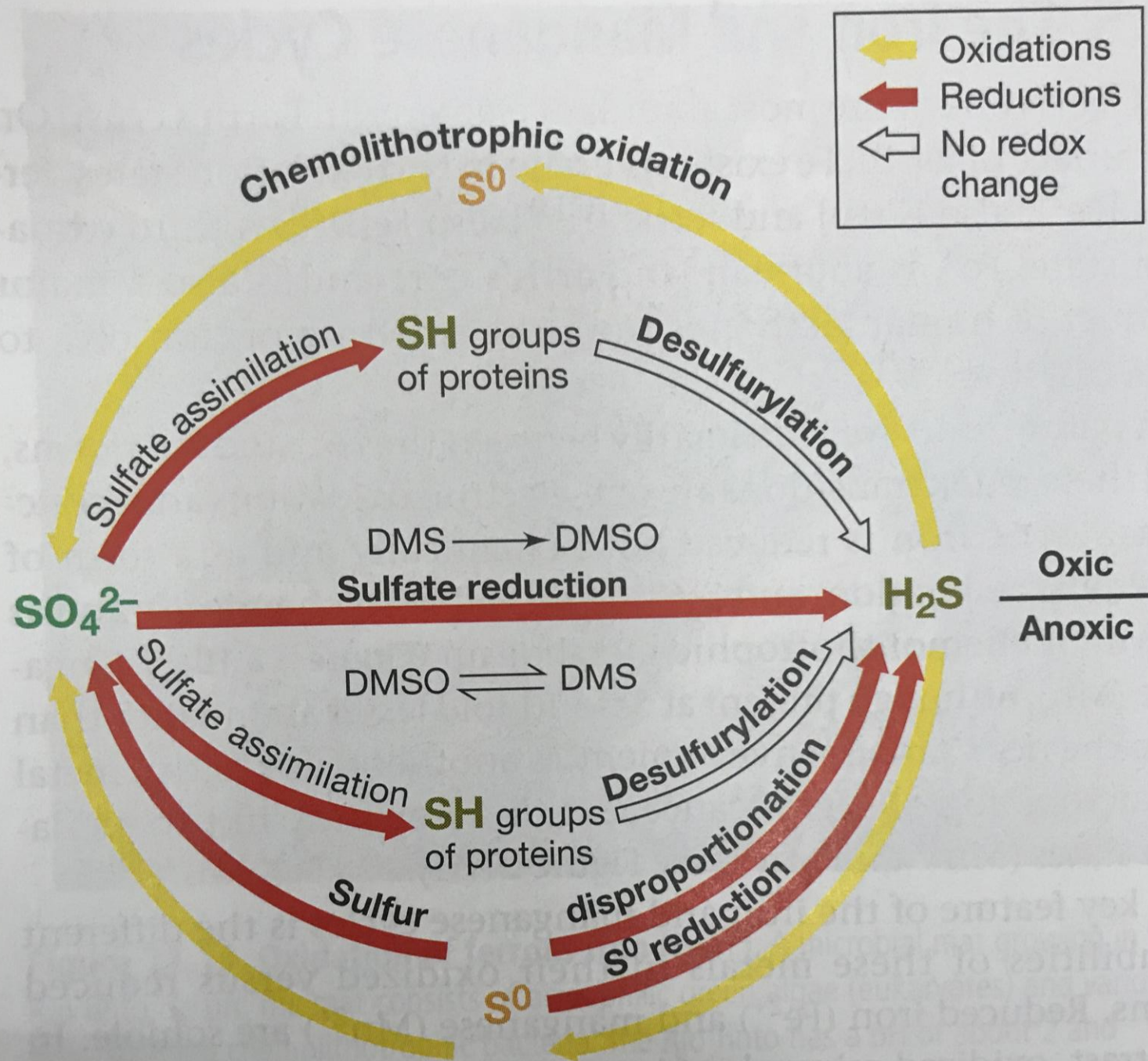
Important 'S' forms and their oxidation states

S^{2-} (R-SH, SH^- , H_2S)
 S^0 and
 SO_4^{2-} are the most significant S forms in nature.



Major processes

- **Geological/Geochemical:** Weathering, erosion, runoff, burial, emissions from volcanoes and hydrothermal vents, abiotic transformations, etc.
- **Atmospheric:** SO_2 chemistry, SO_4 aerosols, transport, deposition, etc.
 - e.g., $n\text{SO}_2 + n\text{O}_2/n\text{H}_2\text{O} \rightarrow n\text{H}_2\text{SO}_4$ (acid rain)
 - DMS/OCS oxidation $\rightarrow \text{SO}_2 \rightarrow \text{SO}_4^{2-}$ (role in cloud condensation)
- **Biological:** Several microbial processes, primary oxidation and reduction reactions.
- **Anthropogenic:** Burning of fossil fuels, Mining, wastewater discharge, etc.



Redox cycle of sulfur

DMS: Dimethyl sulfide

DMSO: Dimethyl sulfoxide

Brock Biology of
Microorganisms, 15th edition

Sulfate reduction

Sulfate-reducing bacteria (SRB)

- Highly diverse and widespread group in nature
- SRB use sulfate as a terminal electron acceptor in respiration
- H₂ or organic matter used as an energy source (limiting factors)



- H₂S is toxic; depending upon pH, H₂S predominates < pH 7, non-volatile SH⁻ and S²⁻ predominant > pH 7 – collectively referred to as sulfide
- Detoxification by combination with Fe to form insoluble compounds (FeS and FeS₂) in most environments - can also combine with Pb, Cd, Zn, etc.

S⁰ reduction



- A form of anaerobic respiration (by some bacteria and hyperthermophilic archaea)
- Same habitats for both groups – leading to H₂S production
- H₂S is also geochemically produced and emitted from sulfide springs, volcanoes, hydrothermal vents, etc.

Mineralization/Desulfurylation of organic S compounds



- Several fungi and bacteria

Sulfide oxidation

- Spontaneous sulfide (abiotic) oxidation predominant process under oxic conditions
- Chemolithotrophic sulfide-oxidizing bacteria (SOB) oxidize H_2S under oxic conditions (also anoxic)



- Anoxygenic phototrophs, e.g., Purple & Green sulfur bacteria, oxidize H_2S under anoxic & light conditions



- **Several sulfide oxidizers can store “S” in cells**

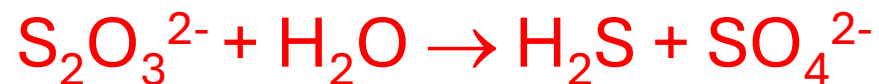
S⁰ oxidation

- S⁰ (chemically stable and insoluble) is oxidized by specific chemolithotrophs, such as *Thiobacillus* and *Acidithiobacillus* sp.



- Lowers the pH of the environment
- Addition of S⁰ can be used to lower the pH of alkaline soils.

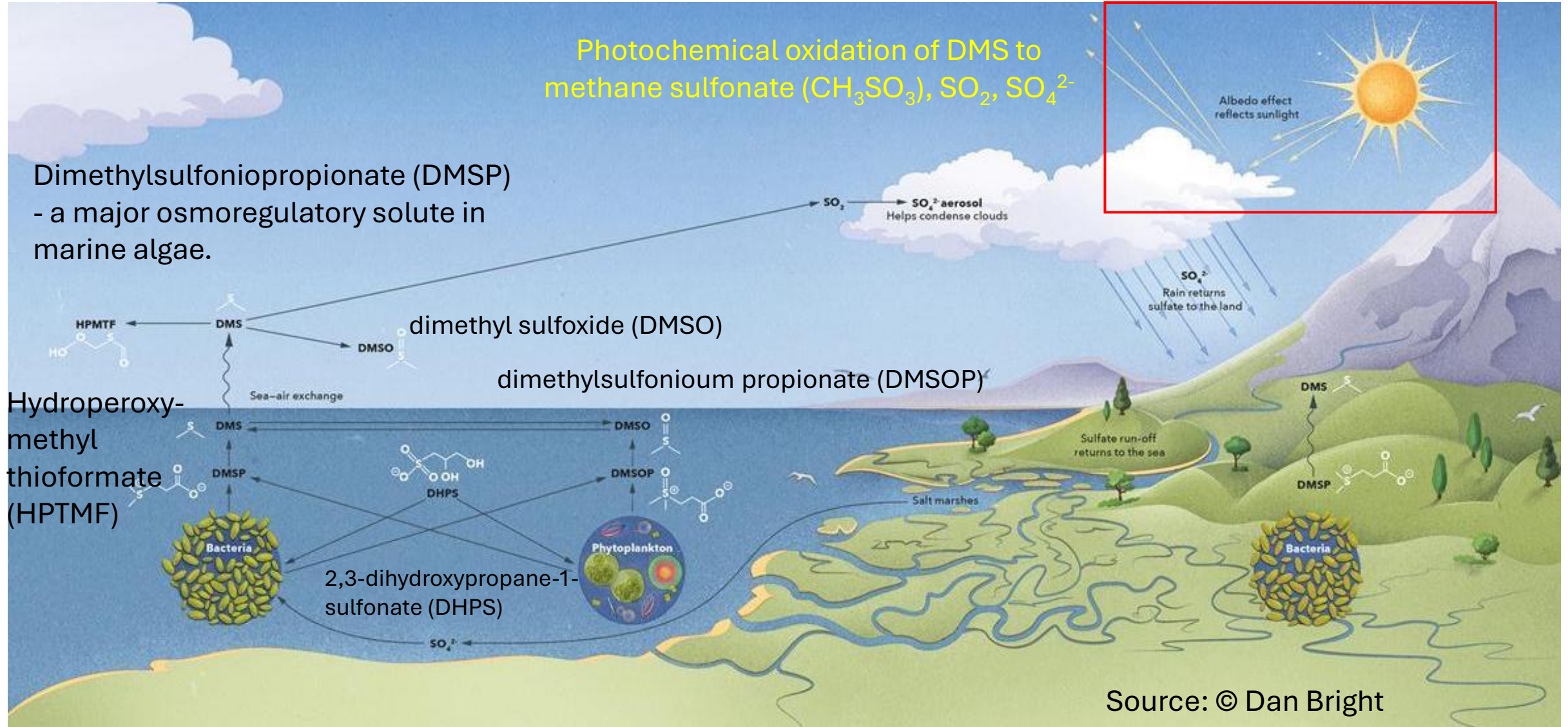
Sulfur disproportionation



- *Desulfovibrio* spp. (e.g., *D. sulfodismutans*)

Cycling of organic 'S' compounds

- Most abundant - dimethyl sulfide ($\text{CH}_3\text{-S-CH}_3$) – DMS



Source: © Dan Bright

- **Microbial DMS transformation (anoxic habitats):**
 - Methanogenesis (CH_4 and H_2S)
 - Electron donor for CO_2 fixation in purple bacteria (dimethyl sulfoxide - DMSO)
 - Electron donor in energy metabolism of some organotrophs & lithotrophs (yielding DMSO)
- DMSO is used as an electron acceptor in anaerobic respiration by some bacteria – producing DMS
- Other organic S compounds include methanethiol (CH_3SH), dimethyl disulfide ($\text{CH}_3\text{-S-S-CH}_3$), and carbon disulfide (CS_2).

TABLE 13.3 Global Budget for Carbonyl Sulfide in the Atmosphere

Sources	COS (10^{12} g S/yr)
Oceans	0.04 ^a
Anoxic soils	0.03
Biomass burning	0.04 ^b
Fossil fuels	0.06
Volcanoes	0.02
Oxidation of CS ₂	0.25 ^c
Oxidation of DMS	0.16
Total sources	0.60
Sinks	
Vegetation uptake	0.24
Soil uptake (oxic)	0.13
Oxidation by OH	0.09
Stratospheric photolysis	0.02
Total sinks	0.48

^a Net.

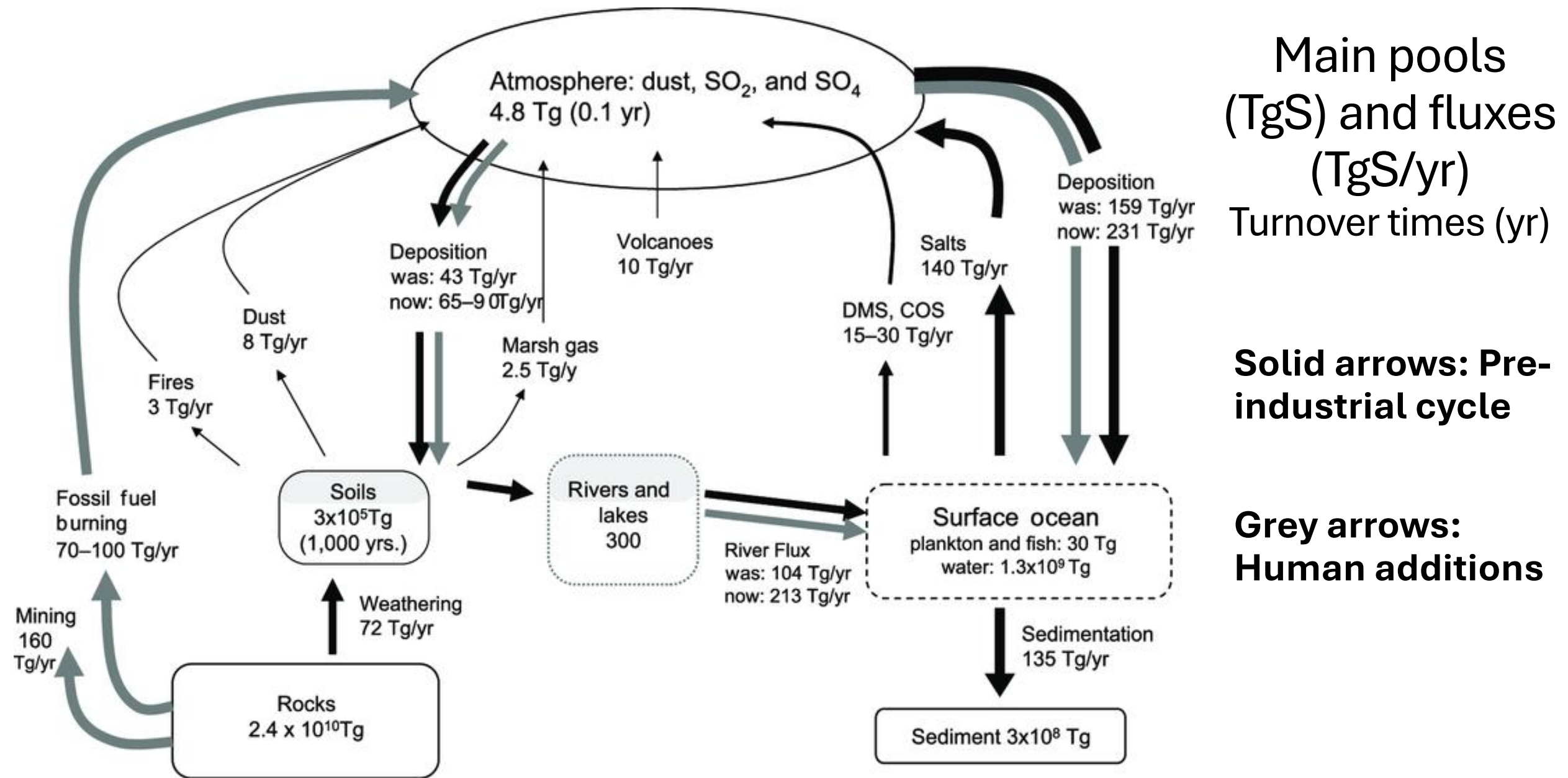
^b Compare, 0.14 Tg S/yr; Andreae and Merlet (2001).

^c 44% from anthropogenic sources.

Sources: From Kettle et al. (2002), except volcanic flux (Chin and Davis 1993).

Carbonyl sulfide (COS) – most abundant S containing gas in the atmosphere

- Average concentration of about 500 ppt
- The residence time for COS in the atmosphere is up to about 5 years.
- Major Sources: Oxidation of carbon disulfide (CS₂), Oxidation of DMS, Fossil fuels, biomass burning, etc.
- **Contributes to the formation of stratospheric sulfate aerosols**



**Main pools
(TgS) and fluxes
(TgS/yr)
Turnover times (yr)**

**Solid arrows: Pre-
industrial cycle**

**Grey arrows:
Human additions**

Impacts of increased S fluxes

- DMS potential effects on climate – sulfate aerosols, Albedo effect, cloud condensation, cooling
- SO₂ in the atmosphere through fossil fuels (and other activities) – reacts with water to form sulfuric acid - **contributes to acidification of rain**
- **Acid rain: Water and soil acidification, Damage to natural and man-made structures**



<https://www.nationalgeographic.com/environment/article/acid-rain>



<https://www.britannica.com/>

- Processing of S-containing minerals for metals – release sulfur in environment
- SO₂ and other S particles– dry deposition – **health impacts – bronchial asthma, respiratory discomfort**
- S-rich waste from mining of coal and metal sulfides – **acid mine drainage (H₂SO₄ rich drainage)**

water contamination

ecological impacts

corrosion to structures

<https://earthworks.org/issues/acid-mine-drainage/>



H₂S release from wetlands and anoxic environments

- **Sulfate reduction** is typically limited by carbon and electron sources - Influx of organic matter when sewage/waste is discharged in water bodies – increase sulfate reduction rates – **more H₂S production**
- H₂S is colorless, flammable, highly toxic gas (leads to rapid CNS, respiratory depression as well as skin and mucous membrane irritant)

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Ludhiana gas leak deaths could have been averted had there been ventilation system along sewer: NGT panel

The report said the people died most probably due to build up of hydrogen sulphide gas inside the sewers, which escaped into the houses through washrooms or toilets

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